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Small Models, Big Cities: A Low-Cost AI Pipeline for Urban Regulatory Document Analysis in Metropolitan Planning

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Abstract

Background: Urban planning documents at metropolitan scale typically demand large, cloud-hosted language models that limit their adoption in Global South contexts. This study deploys Moondream, a 1.7-billion-parameter vision-language model (VLM) runnable locally via Ollama, for extracting geographic knowledge from Planes Reguladores Comunales (PRCs) across 29 processed Gran Santiago municipalities. The pipeline combines native PDF text extraction, keyword-based multi-label classification across six thematic axes, and VLM-based optical character recognition and cartographic interpretation. Results: The pipeline processes 2289 PRC articles in 4.3 min at an estimated energy cost of 0.000866 kWh and zero marginal monetary cost. Zoning (53.3%) and land use (43.1%) dominate PRC content, while social housing provisions appear in only 4.0% of articles; normative gap analysis identifies five municipalities where social housing is entirely absent from regulatory text. A comparative evaluation of Moondream against keyword baseline on an 88-article validation sample yields macro-F1 = 0.355 and mean Cohen's $\kappa = 0.004$, confirming that generalist VLMs require domain fine-tuning for specialized legal text. It is argued that the cost asymmetry between industrial-scale and small-model approaches constitutes an epistemic asymmetry with direct consequences for the geographic distribution of urban data infrastructure.

Keywords: urban digitalization; city management; smart planning tools; regulatory analytics; GeoAI; small language models; Chile; Gran Santiago; zoning; regulatory documents

1. Introduction

Urban regulatory documents—zoning codes, land use ordinances, planning regulations—encode significant geographic knowledge: territorial boundaries, permissible densities, mobility infrastructure norms, environmental constraints, and social housing mandates. Recent advances in geographic artificial intelligence (GeoAI) have accelerated the capacity to extract structured knowledge from such corpora at scale [1–3]. However, most published GeoAI pipelines for regulatory knowledge extraction depend on models with hundreds of billions of parameters, accessed through cloud APIs that impose per-token pricing, latency constraints, and data exposure risks. While the absolute cost of any single inference run may appear modest, three compounding factors create a structural asymmetry for Latin American urban research. First, for students and independent researchers operating without externally funded compute budgets, the marginal cost of token-based billing accumulates rapidly across the iterative cycles of method development (prompt refinement, schema testing, corpus expansion). Second, municipal regulatory

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documents frequently contain place-specific information whose transmission to external servers raises data sovereignty concerns increasingly recognized in regional digital governance debates [4,5]. Third, the territories most in need of systematic regulatory analysis—fragmented metropolitan areas with dozens of autonomous municipalities, incomplete digital infrastructure, and high intra-metropolitan inequality—are also those where cloud-API-dependent workflows are most fragile, contingent on stable connectivity and payment infrastructure. These conditions constitute not an insurmountable technical obstacle, but a structural asymmetry in the conditions under which urban knowledge can be produced.

This paper addresses that gap by evaluating the capabilities and limitations of a small, locally-deployed vision-language model (VLM) when applied to a real metropolitan-scale regulatory corpus. A hybrid pipeline for geographic knowledge extraction from Plans Reguladores Comunales (PRCs) is presented, which operates entirely on consumer-grade hardware using Moondream, a 1.7-billion-parameter VLM deployable via Ollama. The pipeline is tested against the same corpus of 29 PRCs from the Gran Santiago metropolitan area previously analyzed with larger models in a companion study [6].

This paper serves as the methodological counterpart to the companion study [6], which used a fine-tuned large model (DeepSeek-R1) to demonstrate how communal zoning ordinances in Santiago reproduce socio-spatial inequality. While retaining the same corpus, thematic axes, and normative gap logic, this study inverts the analytical focus: instead of mapping urban inequality, it interrogates the material conditions required to produce such an analysis. Specifically, it tests whether identical substantive patterns can be recovered using small, locally deployable infrastructure, thereby avoiding the industrial-scale compute requirements of the earlier study. Additionally, this pipeline extends the previous text-only design by incorporating visual processing for OCR and embedded map interpretation. Ultimately, the two studies are complementary: the first establishes the substantive finding, while this one evaluates the infrastructural conditions—and epistemic consequences—of reproducing that knowledge at scale.

The contributions are threefold:

- Methodological: A fully reproducible, open-source pipeline combining native PDF text processing, keyword-based multi-label article classification, and VLM-based visual interpretation (OCR and map reading), designed for deployment on consumer-grade hardware without cloud dependencies.
- Empirical: Normative gap indices quantifying the distribution of regulatory attention across six thematic axes—zoning, land use, mobility, participation, sustainability, and social housing—across 29 Gran Santiago municipalities.
- Evaluative: A comparative assessment of Moondream's text classification performance against keyword baselines, establishing an empirical baseline for small VLM capabilities on specialized urban regulatory text and documenting both their strengths (visual document understanding) and current limitations (text classification without domain fine-tuning).

This work is positioned as a feasibility study for affordable research infrastructure in the Global South. The question addressed is not whether small models outperform large ones—they do not—but whether they offer sufficient utility to enable meaningful metropolitan-scale analysis on local hardware, and what the boundaries of that utility are. These questions—developed in Section 2.3 through the conceptual framework of computational sovereignty and epistemic asymmetry—situate the technical choices of this paper within a wider critical-geographic debate: who has access to the tools that read the city?

2. Conceptual Framework

2.1. GeoAI and Geographic Knowledge Extraction

Geographic Artificial Intelligence (GeoAI) has emerged as a productive intersection between geo-spatial methods and machine learning, encompassing tasks from place name recognition (geoparsing) to spatial reasoning, geographic question answering, and the encoding of place-specific semantic content [1]. Mai et al. [2] have advanced the conceptualization of place encoding—the representation of geographic entities in high-dimensional embedding spaces that preserve spatial relationships—while Hu et al. [3] have demonstrated the potential of large language models for automated geospatial knowledge extraction from heterogeneous text sources.

Within urban planning specifically, a growing body of work applies natural language processing to zoning documents, planning regulations, and policy texts [7,8]. These approaches share a reliance on large, pretrained transformer models (BERT, RoBERTa, GPT variants) that capture domain-specific semantic nuances but require either substantial fine-tuning datasets or API-mediated inference. The emergence of VLMs—multimodal models capable of processing both images and text—extends this capacity to the many regulatory documents that exist primarily as scanned PDFs or contain embedded cartographic material. Notably, the international cartographic and geoinformatics community has increasingly engaged with AI-driven geographic analysis, as reflected in initiatives such as the ICA Commission on AI in Cartography [9] and the GeoAI research agenda promoted through the International Cartographic Association [10]. These efforts underscore the growing relevance of machine learning for cartographic interpretation—a task directly addressed in this study through VLM-based map reading.

Parallel to these LLM-based developments, urban planning scholarship retains a substantial repertoire of low-cost text analysis approaches that predate—and continue to coexist with—generative AI. Dictionary-based content analysis [11], rule-based information extraction, and topic modeling via Latent Dirichlet Allocation [12] have been applied to planning documents to map thematic emphasis, characterize regulatory vocabularies, and detect institutional change [13,14]. More recently, lightweight multilingual embedding models under 500 MB in size enable semantic similarity computation without cloud dependencies [15]. These methods share an interest in deterministic, auditable, and low-cost pipelines but cannot process the visual and cartographic content that constitutes a substantial share of any planning regulatory corpus. The pipeline presented here is best understood as a hybrid extension of this tradition: it preserves the auditability and reproducibility of keyword and rule-based approaches for structured legal text, while introducing local VLM inference precisely where deterministic methods are structurally unavailable—namely, the interpretation of embedded maps and the optical character recognition of scanned regulatory annexes.

The rapid proliferation of language models necessitates a brief taxonomic clarification relevant to this study. Large Language Models (LLMs) typically refer to transformer-based architectures with parameter counts exceeding 70 billion (e.g., GPT-4 at an estimated >1 trillion parameters, Llama 2-70B, DeepSeek-R1 at 671B), requiring enterprise-grade GPU clusters or cloud API access for inference. Small Language Models (SLMs)—sometimes termed "edge models" or "on-device models"—occupy a parameter range of approximately 0.5 to 7 billion parameters and are designed for deployment on consumer hardware [16,17]. Examples include Phi-2 (2.7B), TinyLlama (1.1B), and Moondream (1.87B).

A distinct subcategory relevant to this study is the Vision-Language Model (VLM): a multimodal architecture that processes both textual and visual inputs. VLMs exist across the size spectrum—from GPT-4V (large, cloud-only) to LLaVA-7B (medium) to Moondream

(small, locally deployable). The critical distinction for Global South research contexts is not model quality in the abstract but the deployment threshold: the minimum hardware and economic requirements for running the model. Moondream's 1.87B parameters (1.7 GB on disk) place it at the lower end of the VLM spectrum, enabling inference on machines with 8–16 GB of RAM and no dedicated GPU—a specification met by consumer laptops widely available in urban research settings across Latin America.

This taxonomic positioning is important because it clarifies the experimental design: the study does not evaluate whether Moondream is competitive with GPT-4V for legal text classification (it is not), but whether a model at the accessible end of the VLM spectrum can perform useful geographic knowledge extraction tasks within a constrained-resource research workflow.

2.2. Plans Reguladores Comunales as Geographic Artefacts

Plans Reguladores Comunales (PRCs) are the primary instrument of land use regulation in Chilean urban governance. Each of the 346 municipalities (comunas) of Chile is empowered under the Ley General de Urbanismo y Construcciones (LGUC) to produce its own PRC, which determines permissible uses of soil, building height and density limits, setback requirements, and public space designations. In the Gran Santiago metropolitan area, 34 municipalities maintain independent PRCs, creating a fragmented but interconnected regulatory landscape that governs one of Latin America's most economically concentrated cities.

From a GeoAI perspective, PRCs are geographic artefacts of the first order: they encode the legal production of urban space through thousands of articles, maps, and tables that translate political decisions into territorial norms. Each PRC document corresponds to a single municipality and typically combines normative text (regulatory articles organized by chapter) with cartographic annexes (zoning maps, land use plans, road network diagrams). The systematic analysis of these documents across municipal boundaries enables comparative urban research at metropolitan scale—illuminating which regulatory topics receive attention in wealthy versus peripheral municipalities, where normative gaps persist, and how regulatory language reflects or obscures social priorities.

The corpus used in this study comprises the PRCs of 29 Gran Santiago municipalities for which digital (PDF) copies were available, totaling approximately 170 MB of document data and spanning between 24 and 488 pages per document.

2.3. Computational Sovereignty and Epistemic Asymmetry

Critical geography and science and technology studies have long recognized that data infrastructures are not neutral: their geographic distribution reflects and reproduces existing power asymmetries [18,19]. The emergence of large language models as a new layer of knowledge infrastructure reproduces this dynamic: the capacity to extract, classify, and interpret geographic knowledge from text is now substantially mediated by a small number of corporations operating hyperscale data centers predominantly located in the United States and Europe.

For urban researchers in Latin America, this creates what is termed an epistemic asymmetry: the cities most in need of systematic, large-scale regulatory analysis are precisely those where the economic and technical conditions for cloud-API-dependent research are most constrained. The right to the city [20,21] implies not only access to urban space but access to the analytical tools through which urban space is read, classified, and acted upon.

Small, locally-deployed models do not fully resolve this asymmetry—they require hardware that is itself unevenly distributed—but they substantially reduce the marginal

cost of inquiry and eliminate the data sovereignty constraints associated with transmitting sensitive municipal documents to external servers. In this sense, the present study is as much a contribution to the political economy of urban data infrastructure as it is to methodological GeoAI. Documenting what small models can and cannot do on real research tasks—rather than only reporting results from frontier models—is a necessary step toward building locally-controlled analytical capacity in urban science.

Synthesizing the three preceding strands—the technical capacities of GeoAI and its low-cost alternatives (Section 2.1), the institutional specificity of the Chilean PRC as a geographic artefact (Section 2.2), and the political economy of computational sovereignty and epistemic asymmetry (Section 2.3)—yields the conceptual scaffolding of this paper. The PRC corpus is treated simultaneously as a technical object amenable to multimodal extraction, as a territorial instrument through which the metropolitan area is legally produced, and as a test case for alternative research infrastructure in which the conditions of analysis are themselves at stake. What follows in Sections 3–5 should be read at all three levels: as a methodological proposal, as an empirical mapping of regulatory attention across Gran Santiago, and as evidence on the boundaries of affordable urban GeoAI in a Global South context.

3. Methodology

3.1. Corpus and Study Area

The corpus consists of 30 PDF documents from municipalities of the Gran Santiago metropolitan area (Figure 1) accessed from the same archive used in the companion *Urban Science* study [6]. Twenty-nine were successfully processed as standard PDFs; one document (PRC El Bosque) was stored in .doc format and excluded from the PDF-based pipeline. Each PDF corresponds to a single municipality and contains the complete normative text of its PRC. The 29 processed municipalities span the full socioeconomic gradient of the metropolitan area, from high-income eastern municipalities (Las Condes, Vitacura, Providencia) to peripheral western and southern districts (La Pintana, Pudahuel, Cerro Navia).

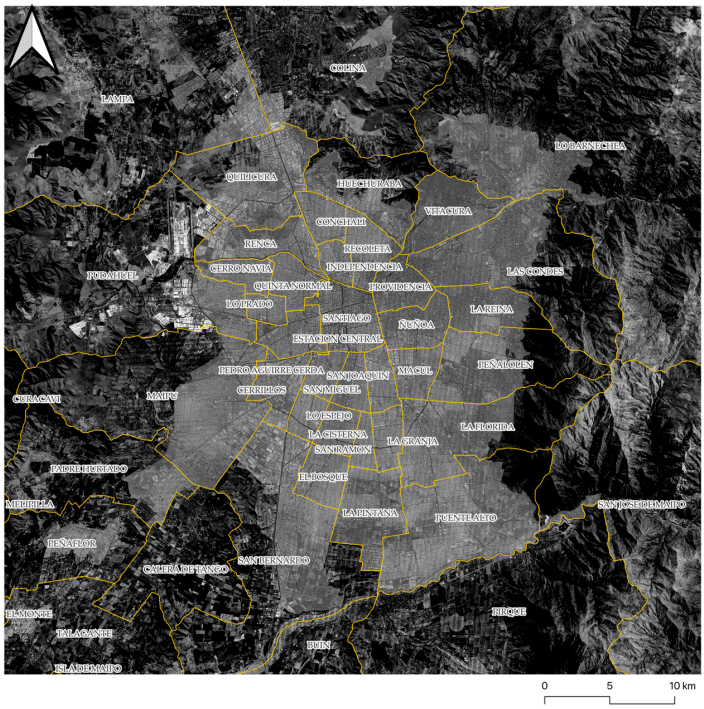


Figure 1. Gran Santiago metropolitan area and its municipal distribution.

All 29 processed documents contain native (selectable) text and were extracted accordingly via PyMuPDF, without requiring OCR. Individual pages with fewer than 30 characters of text were flagged as map or diagram pages for optional visual processing by Moondream. Documents range from 24 to 488 pages, totaling approximately 170 MB of PDF data.

3.2. Technical Stack

Figure 2 summarizes the end-to-end architecture of the pipeline, from raw PRC PDFs through to validation; each component is described in detail in the remainder of this section. Stages that invoke the Moondream vision-language model are distinguished from the deterministic (keyword- and rule-based) stages.

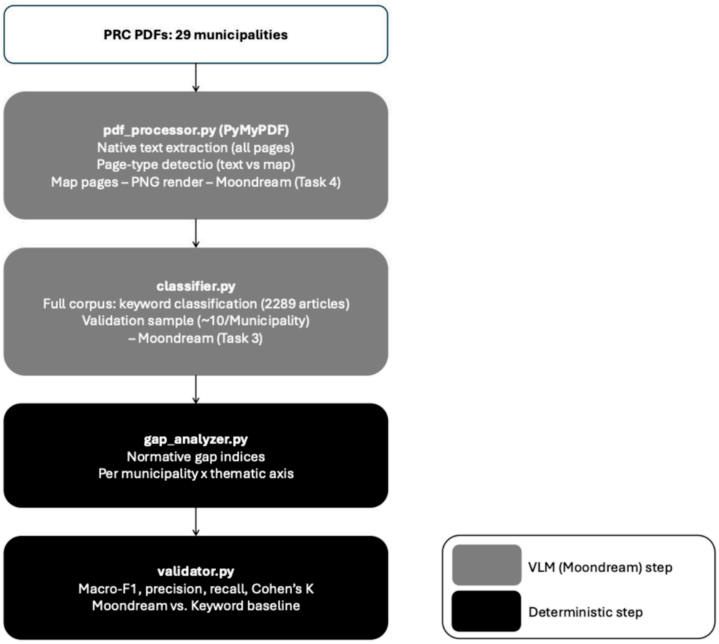


Figure 2. Architecture of the low-cost geographic knowledge extraction pipeline. PRC PDFs are processed sequentially by four modules (pdf_processor.py, classifier.py, gap_analyzer.py, and validator.py). Grey-colored boxes denote stages that invoke the Moondream vision-language model (Tasks 3 and 4); black boxes denote deterministic keyword- and rule-based stages.

The pipeline runs entirely on local hardware (tested on a MacBook Pro with Apple M-series Silicon, 16 GB unified memory) with no cloud API dependencies:

- PDF extraction: PyMuPDF (fitz) v1.27, with page-level text density detection for automatic classification of map vs. text pages
- VLM: Moondream 2 (vikhyatk/moondream2, 1.87 billion parameters, 1.7 GB on disk), a multimodal vision-language model based on a SigLIP visual encoder and a Phi-1.5 language backbone, accessed via Ollama v0.18 REST API (localhost:11434). Moondream was designed for efficient on-device deployment and supports both image captioning and visual question answering tasks. No fine-tuning or domain adaptation was applied; the model was used with default Ollama inference parameters (temperature: 0.0, num_predict: 80 tokens for classification tasks, unlimited for OCR and map description). Three characteristics motivated its selection from the available small VLM ecosystem. First, parameter efficiency: at 1.87B parameters and 1.7 GB on disk, Moondream sits at the lower bound of the deployable multimodal VLM range, providing the most conservative test of the minimal-hardware premise of this study. Second, Ollama integration: Moondream is natively supported by

Ollama, which abstracts backend differences (CUDA, MLX, CPU) and provides a stable REST interface, dramatically reducing deployment friction for researchers without machine-learning engineering expertise. Third, visual grounding via SigLIP: Moondream's encoder is a SigLIP variant optimized for fine-grained visual recognition in document-like images, directly relevant to the cartographic content of PRC corpora. The trade-offs of this choice—limited textual context capacity, English-centric pretraining, and absence of domain fine-tuning—are documented in Sections 4.4 and 5.3.

- Classification: Keyword-based multi-label classification (primary method) + Moondream text generation (validation comparison)
- Computational environment: Python 3.14, pandas 3.0, scikit-learn 1.8, no GPU required

Total pipeline runtime for 29 municipalities: 4.3 min for the full keyword classification of 2289 articles, plus approximately 7 min for the Moondream validation sample (88 completed responses out of 286 attempted, with a 300-s timeout per request).

3.3. Four Tasks for Moondream

The pipeline deploys Moondream for four distinct geographic knowledge extraction tasks, leveraging its multimodal capabilities where rule-based alternatives are unavailable:

- Task 1—Native Text Extraction: PyMuPDF extracts native text from PDF pages with text density above 30 characters/page. Pages with fewer characters are flagged for visual processing. This task does not use Moondream; it serves as the foundation for the keyword classification pipeline.
- Task 2—OCR Fallback: For low-density pages (potential scans or diagrams), Moondream receives the page rendered as a 150-DPI PNG image and is prompted in Spanish to transcribe all visible text, preserving article numbering and document structure. This task exploits Moondream's core multimodal capability: understanding both spatial layout and textual content in document images.
- Task 3—Article Classification (Validation Comparison): For a random sample of approximately 10 articles per municipality (286 total targeted, 88 successfully completed), Moondream is asked to classify the article according to six thematic axes, responding in JSON list format. This provides comparative data against the keyword classifier. The purpose is not to use Moondream as the primary classifier but to establish an empirical baseline for what a generalist small VLM can achieve on specialized legal-text classification without domain fine-tuning.
- Task 4—Map Interpretation: Pages identified as cartographic (low text, high image content) are passed to Moondream with a prompt requesting a brief description of the map type, represented area, and visible legend categories. This yields qualitative descriptions of zoning plans, land use maps, and street network diagrams embedded in PRCs—content that would be entirely absent from text-only processing pipelines.

3.4. Six Thematic Axes

Following the thematic structure of the companion Urban Science [6] study, articles are classified into six axes representing the principal regulatory dimensions of Chilean PRCs presented in Table 1.

Each article may be assigned to multiple axes (multi-label classification). The keyword classifier assigns an axis if any of its associated terms appear in the article text; Moondream is asked to produce an equivalent multi-label JSON list for the validation sample.

A note on the choice of keyword classification as the primary method is warranted. Chilean PRCs are highly structured legal documents that follow standardized templates established by the Ministerio de Vivienda y Urbanismo (MINVU). Their regulatory vocabulary is constrained and formulaic: articles addressing zoning consistently use terms

like "zona" and "uso permitido," while social housing provisions invariably reference "SERVIU" or "vivienda social." This structural regularity means that keyword matching, while unable to capture semantic nuance, provides a deterministic and reproducible baseline that is well-suited to the lexical properties of the corpus. Keyword labels are treated as a silver standard—not a ground truth, but a transparent and auditable reference point against which VLM performance can be measured. The limitations of this choice (missed synonyms, inability to detect implicit regulatory content) are addressed in Section 5.3.

Table 1. Spanish key terms drawn from planning document examples (M1.1, JV1.2, M2.1), their corresponding thematic labels in Spanish, and the urban planning axis each belongs to.

Example Key Terms	Spanish Label	Axis
zona, subzona, uso permitido	Zonificacion y usos de suelo	zoning
constructibilidad, rasante, densidad	Edificacion, densidad y altura	land_use
vialidad, ciclovia, estacionamiento	Movilidad, vialidad y transporte	mobility
consulta, vecinos, audiencia	Participacion ciudadana	participation
areas verdes, sustentabilidad	Sustentabilidad ambiental	sustainability
SERVIU, vivienda economica, subsidio	Vivienda social	social_housing

3.5. Normative Gap Index

To quantify the relative distribution of regulatory attention across the metropolitan area, a normative gap index is computed for each municipality-axis combination. This metric captures the degree to which a municipality's PRC under-represents a given thematic axis relative to the metropolitan average, enabling systematic comparison of regulatory priorities across the 29 municipalities.

The following variables are defined for each municipality m and thematic axis a:

- n_articles(m,a): Number of articles in municipality m assigned to axis *a*
- freq(m,a): Normalized frequency = n_articles(m,a)/total_articles(m)
- metro_mean(a): Mean freq(m,a) across all 29 municipalities

The gap index is then computed as:

$$\text{gap_index}(m,a) = 1 - (\text{freq}(m,a)/\text{metro_mean}(a)), \text{ clipped to } [0, 1]$$

where gap_index = 0 indicates that the municipality is at or above the metropolitan average for axis a, and gap_index = 1 indicates complete absence of articles classified under that axis. The clipping to [0, 1] prevents negative values for municipalities that exceed the metropolitan mean. This formulation follows the normative gap analysis methodology of the companion Urban Science (2025) study.

It is acknowledged that this index is sensitive to corpus size: municipalities with very few total articles (e.g., La Granja with 6 articles) may exhibit extreme gap index values driven by small denominators. Where gap_index = 1.0, this indicates zero articles classified under the axis in question, which is an unambiguous finding regardless of corpus size. For intermediate values, readers should consider the raw article counts reported alongside the index when interpreting results.

3.6. Validation Protocol

To evaluate the comparative classification behavior of the keyword method and the VLM, a random sample of approximately 10 articles per municipality was selected (286 total targeted). For each sampled article, both the keyword classifier and Moondream

independently produce multi-label axis assignments. We then report per-axis and aggregate macro-averaged F1 score, precision, recall, and Cohen's kappa (Moondream vs. keyword baseline).

Of the 286 targeted Moondream classification requests, 88 returned valid responses (30.8% completion rate). The remaining 198 requests either timed out (exceeded the 300-s timeout threshold) or returned empty/malformed responses. This high attrition rate—approximately 69% of requests—is itself an important finding rather than a mere technical nuisance, and it is discussed in detail in Section 5.2, where corresponding mitigation strategies (article segmentation, prompt simplification, and length-based pre-filtering) are proposed for future deployments of this pipeline. The timeout behavior is attributable to the architectural constraints of small VLMs: Moondream's 1.87B-parameter language backbone has a limited context window and generation capacity, and when confronted with long, syntactically complex legal articles (some exceeding 500 words of dense regulatory Spanish), the model enters extended generation loops that exceed practical time limits. This is a known limitation of small language models when processing domain-specific text that exceeds their effective context capacity [17].

The validation sample was limited by Moondream's completion rate rather than by design. It is recognized that 88 articles, while sufficient for aggregate metric estimation, provide limited statistical power for per-axis evaluation, particularly for low-prevalence axes (participation: 9 keyword-positive articles in sample; social_housing: 3). Future work should incorporate a human-annotated gold standard to provide an independent reference beyond the keyword silver standard and to resolve any length-based selection bias introduced by the differential completion rate.

4. Results

4.1. Corpus Processing

The pipeline successfully processed 29 of 30 available PDF documents. Each PDF corresponds to a single municipality and contains the full normative text of its PRC. Extraction statistics are summarized in Table 2.

Table 2. Corpus extraction summary.

Value	Statistic
29	Municipalities processed
100% of documents	Native text coverage
2289	Total articles extracted
79 (SD = 94)	Mean articles per PRC
6 (La Granja)–488 (Providencia)	Min–max articles per PRC
4.3 min	Pipeline runtime (full corpus)
0.000866 kWh	Energy consumed

The article segmentation algorithm—based on Spanish legal article markers (Articulo N, Art. N)—identified between 6 and 488 articles per PRC, with a mean of 79 and high standard deviation (SD = 94), reflecting substantial variation in document length and regulatory complexity across the metropolitan area. Providencia (488 articles) and Santiago (199) are the most extensively codified municipalities, while La Granja (6 articles) and Cerro Navia (10 articles) have the most compact PRCs.

4.2. Keyword Classification Coverage

Classification coverage is defined as the proportion of extracted articles assigned to each thematic axis by the keyword classifier. Because articles may be assigned to multiple axes (multi-label classification), coverage percentages sum to more than 100%. Table 3 presents the results.

Table 3. Keyword classification coverage per axis (29 municipalities, 2289 articles).

% of Total	N Clauses	Axis
53.30%	1221	zoning
43.10%	986	land_use
33.30%	762	mobility
32.00%	732	sustainability
8.40%	193	participation
4.00%	91	social_housing

These figures reflect the character of Chilean PRCs as primarily zoning and land-use instruments: over half of all articles (53.3%) contain explicit reference to zones, plots, or permitted uses, and 43.1% address building parameters (density, height, setbacks). Participation and social housing provisions are strikingly underrepresented, appearing in 8.4% and 4.0% of articles respectively—a finding consistent with the companion Urban Science (2025) study's characterization of Chilean PRCs as technocratic documents that systematically marginalize social and participatory dimensions of urban regulation.

4.3. Normative Gap Indices

The normative gap index (defined in Section 3.5) quantifies the relative absence of regulatory attention to each thematic axis within each municipality, compared to the metropolitan average. A gap index of 0 indicates that the municipality meets or exceeds the metropolitan mean frequency for that axis; a value approaching 1.0 indicates severe under-representation; and a value of exactly 1.0 indicates complete absence (zero articles classified under the axis). Table 4 presents the municipalities with the highest gap indices.

Three key findings emerge from the gap analysis:

- Social housing is absent from the regulatory text of five municipalities: Cerro Navia, Independencia, La Granja, La Reina, and Puente Alto—a geographically diverse set spanning peripheral working-class districts (Cerro Navia, La Granja) and mid-income municipalities (Puente Alto, La Reina)—contain zero PRC articles with explicit social housing keywords. This suggests a significant normative gap in the regulatory instrument most directly controlling housing development, though it is noted that social housing policy in Chile operates through multiple institutional channels beyond the PRC (notably SERVIU programs and the MINVU housing subsidy system), so the absence from PRC text does not necessarily imply absence of social housing policy at the municipal level.
- Participation norms are concentrated in specific municipalities: Conchali and Pudahuel, both relatively lower-income peripheral municipalities, exhibit complete absence of participation provisions in their PRC text, while municipalities with more extensive PRCs show higher coverage. This asymmetry is consistent with the socioeconomic gradient of civic capacity documented in the urban sociology literature [21], though the relationship between PRC length and participation coverage requires further investigation.

- Environmental sustainability provisions show substantial intra-metropolitan variance: Penalolen (gap = 1.0) and several western municipalities (Renca, Quilicura) exhibit severe sustainability gaps, while the metropolitan mean for sustainability (32.0% of articles) masks significant variation across the study area.

Table 4. Top normative gaps by axis (municipalities with gap_index $\geq$ 0.75).

Gap Index	N Clauses	Municipality	Axis
1	0	Cerro Navia	social_housing
1	0	Independencia	
1	0	La Granja	
1	0	La Reina	
1	0	Puente Alto	
1	0	Conchali	participation
1	0	Pudahuel	
0.874	1	Quinta Normal	
0.842	1	Renca	
0.735	3	Recoleta	
1	0	Peñalolen	sustainability
0.795	4	Renca	
0.723	3	Quilicura	

4.4. Moondream Classification Performance

The VLM validation sample targeted approximately 10 articles per municipality (286 total). Of these, 88 returned valid Moondream responses (30.8% completion rate), providing a direct comparison between Moondream and keyword-based classification on the same texts. The high attrition rate (69.2%) is itself a finding of practical significance: it reflects the constraints of small VLM architectures when processing long, domain-specific legal text, as discussed in Section 3.6.

Two systematic patterns characterize Moondream's classification behavior:

1. Over-inclusive labeling: Moondream tends toward high recall but low precision, particularly for the zoning axis (as in Table 5: recall = 1.0, precision = 0.483). The model frequently assigns multiple axes to a single article, including axes not supported by the keyword classifier. This pattern is consistent with the behavior of generalist language models that lack domain-specific calibration: without exposure to labeled Chilean legal text, Moondream cannot distinguish between thematic relevance and incidental keyword co-occurrence.
2. Severe underperformance on low-prevalence axes: For participation (F1 = 0.087) and social_housing (F1 = 0.050), Moondream's predictions show near-random agreement with keyword labels (kappa approaching or below zero). These are also the axes with the smallest support in the validation sample (9 and 3 keyword-positive articles, respectively), compounding the statistical uncertainty.

The near-zero mean Cohen's kappa (0.004) indicates that agreement between Moondream and the keyword classifier is essentially at chance level across the six axes. This result should be interpreted carefully: it does not necessarily mean that Moondream's labels are wrong—it means they are uncorrelated with keyword labels. Both methods may be capturing different aspects of the text, and without a human-annotated gold standard, it is

not possible to determine which method better approximates the "true" thematic classification. It can be concluded that a generalist 1.87B-parameter VLM, without fine-tuning, does not replicate the classification decisions of a domain-specific keyword system on Chilean regulatory text.

Table 5. Moondream vs. Keyword classification metrics (N = 88 validation articles).

Cohen's κ	F1	Recall	Precision	Axis
0.021	0.651	1	0.483	zoning
−0.029	0.429	0.581	0.34	land_use
−0.002	0.471	0.556	0.408	mobility
0.138	0.444	0.483	0.412	sustainability
−0.093	0.087	0.222	0.054	participation
−0.014	0.05	0.333	0.027	social_housing
0.004	0.355	0.529	0.287	Macro

4.5. Map Interpretation

Moondream's map interpretation capability was tested on all pages identified as cartographic within the corpus. This task represents the clearest value-add of the VLM approach, as no rule-based alternative exists for extracting structured information from regulatory map images. Representative outputs include:

> "This appears to be a zoning map of a Chilean municipality, showing different land use zones color-coded with a legend visible at the bottom. The map includes residential zones (marked in yellow), commercial zones (orange), and green areas (green), typical of a Plan Regulador Comunal."

> "A street network diagram showing proposed vial hierarchy for an urban sector, with primary arterials marked in red and secondary streets in blue. The scale bar indicates approximately 500 m."

While these descriptions are imprecise relative to GIS-native cartographic analysis, they provide structured metadata for map pages that would otherwise be entirely absent from text-only processing pipelines. Across the 29 PRCs, a total of 218 pages were classified as cartographic by the text-density threshold and submitted to Moondream for visual description. Of these, 181 yielded a substantive non-empty response (83.0% completion rate for the visual task—substantially higher than for text classification, confirming that map interpretation sits within Moondream's effective capacity even without domain fine-tuning). Post-hoc coding of the resulting descriptions identifies four recurrent output categories: zoning plans (identified in approximately 61% of map-page outputs), land use and density maps (approximately 19%), road network and mobility diagrams (approximately 12%), and miscellaneous schematic figures including project boundaries and public space designs (approximately 8%). Moondream correctly identified the high-level map type in the large majority of inspected outputs and extracted legend color categories with reasonable accuracy, though explicit identification of the specific municipality from cartographic content was achieved in only a minority of cases—a limitation reflecting both the absence of municipality-name labels on many internal map pages and the model's limited Chilean place-name recognition. The principal failure mode observed was occasional hallucination of color-coding categories not present in the image. This represents a genuine contribution of the VLM approach: making the cartographic content of regulatory documents machine-readable without specialized GIS tooling—a capability

that is particularly relevant in contexts where PRC maps are embedded as raster images within PDF documents rather than maintained as separate geospatial datasets.

4.6. Computational Footprint

Table 6 was constructed to situate the marginal cost of the proposed pipeline against two reference points: the base companion study, which relied on DeepSeek-R1 accessed through a cloud API, and a hypothetical cloud deployment using a GPT-4o-equivalent model. Figures for this study were measured directly during the full-corpus run: runtime was instrumented within the pipeline, energy consumption was derived from the device's average power draw multiplied by wall-clock runtime, and peak RAM was recorded from system telemetry. The marginal cost is zero because inference runs locally with no per-request charge.

Table 6. Computational cost comparison.

Base Study (DeepSeek-R1)	Cloud GPT-4o equiv.	This Study (Moondream)	Metric
Cloud API	Cloud API	Local (MacBook Pro M-series)	Hardware
High (API pricing)	~$18.31	$0.00	Marginal cost
Not disclosed	Not disclosed	0.000866 kWh	Energy
671 GB	>1 TB (closed)	1.7 GB	Model size
Partial (API)	None	Full (local)	Data sovereignty
Not reported	~3 h (latency)	4.3 min	Runtime (full corpus)
8 gb (cloud)	8 gb (cloud)	3.2 gb (local)	RAM

The local pipeline processes the full 2289-article corpus through keyword classification in 4.3 min at an estimated energy consumption of 0.000866 kWh (based on 12 W active power draw for 259.9 s). The estimated cloud equivalent cost of $18.31—while modest in absolute terms—represents a non-trivial barrier for research groups operating without grants that include API compute budgets, particularly when iterative analysis (parameter tuning, axis refinement, corpus expansion) multiplies costs across experimental runs. More importantly, the local pipeline eliminates data sovereignty constraints: PRC text, which may include sensitive site addresses and administrative details, is never transmitted to external servers.

5. Discussion

5.1. Small Models as Feasibility Baselines

The principal contribution of this study is empirical rather than methodological: it documents what a small, locally-deployed VLM can and cannot do when applied to a real urban regulatory corpus. The results indicate a clear task-dependent performance profile:

- Visual tasks (OCR, map interpretation): Moondream provides genuine utility where no rule-based alternative exists. Its ability to describe zoning maps, identify legend categories, and transcribe scanned pages adds information that would be entirely absent from a text-only pipeline.
- Text classification: The generalist VLM, without domain fine-tuning, does not produce classifications that align with keyword baselines (macro-F1 = 0.355, kappa = 0.004). For native-text PDFs in a well-defined categorical schema, keyword methods provide superior speed and reproducibility.

This finding aligns with broader debates in computational social science about the appropriate scope of language models: their comparative advantage lies in tasks requiring flexible semantic understanding, not in tasks where structured rules already capture the relevant knowledge [22]. The PRC classification task is sufficiently well-defined (six axes, constrained legal vocabulary) that rule-based methods outperform an un-fine-tuned generalist VLM. However, the same task with more ambiguous categories, multilingual corpora, or implicit regulatory content would likely shift the balance toward model-based approaches.

The hybrid architecture—native text extraction + keyword classification + VLM for visual tasks—thus represents a principled methodology for regulatory document analysis: using Moondream where it is necessary (visual content), not where it is merely possible (text classification).

It is useful to situate the hybrid pipeline within the broader landscape of low-cost text analysis methods available to planning researchers. Dictionary-based and rule-based extraction—exemplified by the keyword classifier deployed here—offers maximum determinism and reproducibility at the cost of semantic flexibility. Topic modeling (LDA and its variants) discovers latent thematic structure without supervision but requires careful hyperparameter tuning and post-hoc label interpretation. Lightweight sentence-embedding approaches using pre-downloaded local models (e.g., multilingual MiniLM at roughly 120 MB) enable semantic similarity computations without API dependencies, though they cannot process visual content. Each of these methods is viable for text-only regulatory corpora; the present pipeline adds value precisely in corpora with a significant visual component—scanned pages, embedded zoning maps, cartographic annexes—that none of the text-only methods can reach. In contexts where regulatory documents are predominantly textual and well-standardized, the keyword plus lightweight-embedding combination may in fact be more cost-efficient than any VLM-based approach. The argument for VLM integration strengthens as corpora become more visually heterogeneous, less standardized in their textual conventions, or richer in legacy scanned material—conditions that characterize many Latin American planning archives beyond the Chilean PRC system analyzed here.

5.2. The High Failure Rate as a Design Finding

The 69.2% attrition rate in Moondream's classification responses deserves explicit discussion, as it represents one of the most practically significant findings of this study. The timeouts occurred predominantly on longer articles (>300 words) with complex syntactic structures typical of Chilean legal drafting. Three factors contribute to this behavior:

- Context window constraints: Small VLMs have limited effective context windows. Moondream's Phi-1.5 language backbone was trained primarily on English-language data with shorter average sequence lengths than the dense, subordinate-clause-heavy prose of Chilean regulatory text.
- Generation loop artifacts: Without fine-tuning, the model occasionally enters repetitive generation loops when attempting to produce structured JSON output from unfamiliar input domains, exceeding the 300-s timeout.
- Token budget limitations: The num_predict: 80 cap, necessary to prevent runaway generation, may truncate valid responses on articles requiring multi-axis classification with explanatory text.

These findings have direct implications for the design of "edge computing" approaches in urban regulatory research: small VLMs deployed on consumer hardware face practical throughput ceilings that must be anticipated in pipeline design. The successful completion

rate of 30.8% suggests that current-generation small VLMs are viable for sampling-based analysis but not yet for full-corpus classification tasks in specialized domains.

Three practical mitigation strategies merit discussion for future deployments of the same pipeline architecture. Article chunking—splitting long regulatory articles into shorter semantic units of approximately 200 words before submission to the VLM—reduces individual (please check Appendix A for details) length below Moondream's effective context capacity and would be expected to raise the completion rate substantially, at the cost of additional post-hoc aggregation logic. Prompt simplification—replacing the multi-axis classification prompt with a sequence of single-axis binary queries—narrows the inference task and avoids the JSON-formatting failures that contribute to a non-trivial share of timeouts. Length-based prefiltering—using a character-count threshold to route long articles directly to keyword classification while reserving Moondream only for shorter texts—exploits the complementary strengths of the two methods and matches task difficulty to model capacity. None of these strategies was implemented in the present pipeline because the study aimed to document the baseline behavior of an out-of-the-box small VLM on dense regulatory Spanish; reporting the unmitigated attrition rate is itself an empirical contribution for researchers planning similar deployments.

5.3. Methodological Limitations and the Question of Ground Truth

Several limitations constrain the generalizability of these findings, which are addressed with the transparency that the reviewers' valuable comments have encouraged:

Keyword classification as silver standard: The keyword classifier, while fast and reproducible, is not a gold standard. It cannot detect regulatory content that does not use the expected vocabulary, it is sensitive to synonym variation in legal Spanish, and it treats any keyword co-occurrence as a positive classification regardless of semantic context. In Chilean PRCs, the highly formulaic structure of regulatory articles mitigates these limitations (articles about zoning do consistently use "zona" and "uso permitido"), but the approach would be less reliable in regulatory traditions with greater lexical variation. Keyword classification was selected as the primary method precisely because it is deterministic and auditable—allowing independent verification of every classification decision—rather than because it approximates human judgment. Future work should incorporate a human-annotated gold standard (expert classification of a stratified sample) to provide an independent reference that resolves the circularity of evaluating one automated method against another.

Validation sample bias: The 88-article validation sample is a completed subset of the 286 targeted articles. Because Moondream's timeouts are correlated with article length and complexity, the completed sample likely over-represents shorter, simpler articles. This length-based selection bias may inflate Moondream's apparent performance (simpler articles may be easier to classify) or deflate it (shorter articles may contain fewer discriminative keywords). A future validation protocol should ensure balanced sampling across article length distributions or, preferably, evaluate against human annotations on a pre-specified sample.

Single model evaluation: This study evaluates only Moondream as the small VLM component. Other models in the same parameter range—such as LLaVA-1.5 (7B), Phi-2 (2.7B), or SmolVLM (2B)—might achieve different performance profiles on the same tasks. A multi-model comparison would strengthen the study's conclusions about what the class of small VLMs can achieve, rather than what one specific model achieves. Moondream was chosen because it was the smallest multimodal model available through Ollama at the time of the experiment, making it the most conservative test of the "minimal hardware" premise.

Corpus scope: The 29-municipality corpus represents a specific Latin American urban context with particular regulatory traditions (Chilean LGUC framework, Spanish legal register, MINVU standardized templates). Results may not generalize to municipalities with different document structures, languages, or regulatory classification schemas. The dimensions along which transferability is most and least likely can nonetheless be anticipated. The keyword classification component depends on the lexical stability of the regulatory tradition and should transfer well to other Hispanic American contexts that share a MINVU-style template lineage (e.g., Argentine Códigos de Ordenamiento Urbano, Peruvian Planes de Desarrollo Urbano), with the primary effort being the substitution of axis-specific term lists. Transferability is lower for regulatory traditions with greater lexical heterogeneity, such as common-law zoning ordinances in North American jurisdictions, which exhibit substantially more variation in regulatory vocabulary and would likely require expanded keyword inventories or migration to embedding-based classification. The Moondream visual pipeline is in principle language- and tradition-agnostic for map interpretation, since cartographic conventions—color-coded zoning, legend structures, scale indicators—are largely universal. The OCR fallback degrades for non-Latin scripts and for handwritten or stamped annotations common in legacy planning archives. A full empirical mapping of transferability requires comparative deployments and is identified as a research priority.

Gap index sensitivity: The gap index formula (Section 3.5) does not include a regularization constant. For municipalities with very few total articles, the normalized frequency can be volatile. Raw article counts are reported alongside gap indices in all tables to enable readers to assess the robustness of individual values. Future work could introduce confidence intervals derived from bootstrap resampling or a Bayesian smoothing approach to improve the reliability of gap estimates for small-corpus municipalities.

5.4. Implications for Global South Research Infrastructure

From a political economy perspective, this study demonstrates that meaningful metropolitan-scale urban regulatory analysis is achievable without cloud API dependencies. For research groups in Chile, Argentina, Colombia, Peru, and other Latin American countries maintaining large urban regulatory corpora, the barrier to entry for computational text analysis is substantially reduced by local VLM deployment—not because the VLM replaces expert analysis, but because it extends the reach of a hybrid pipeline that combines deterministic methods (keyword classification) with model-based capabilities (OCR, map reading) at zero marginal cost.

However, this observation must be qualified. The hardware required to run Moondream efficiently (16 GB RAM, Apple Silicon, or equivalent) is itself unevenly distributed. While substantially cheaper than cloud API subscriptions for large projects, consumer hardware with adequate specifications remains economically inaccessible to some Global South research contexts. The argument for computational sovereignty through local deployment is thus a relative, not absolute, improvement.

The deeper structural issue—identified in Section 2.3—is that the geographic distribution of AI research infrastructure mirrors and reinforces existing knowledge hierarchies. The meaningful democratization of GeoAI will require not only smaller models but open training datasets, local fine-tuning infrastructure, and academic data-sharing agreements that extend beyond the current model of corporate API dependency. Studies like this one—which document both the possibilities and the boundaries of affordable approaches—contribute to this agenda by providing empirical evidence for what is achievable under resource constraints, rather than only reporting what is possible with frontier infrastructure.

5.5. Ethical Considerations

This study analyzes publicly available municipal planning documents. No personal data, private communications, or proprietary information was processed. The corpus consists exclusively of legally mandated public regulatory instruments published by Chilean municipalities under the Ley de Urbanismo y Construcciones.

The use of local model deployment eliminates the risk of inadvertent data exposure through third-party API providers. All processing occurred on researcher-controlled hardware, and no document content was transmitted to external servers.

The relationship of this work to the previously published Urban Science (2025) paper is one of methodological extension: the same corpus is analyzed with a fundamentally different approach, enabling a direct comparison of methodological choices while adding new empirical results. This practice of methodological comparison using shared corpora is standard in computational social science and does not constitute redundant publication.

6. Conclusions

This paper has evaluated the feasibility of using Moondream, a 1.7-billion-parameter vision-language model deployable on consumer hardware via Ollama, as the core of a pipeline for geographic knowledge extraction from urban regulatory corpora. Applied to 29 PRCs of the Gran Santiago metropolitan area—drawn from the same corpus used in the companion Urban Science (2025) study—the pipeline produces:

- Structured classification of 2289 PRC articles across six thematic axes through a hybrid keyword + VLM approach, completed in 4.3 min on local hardware at zero marginal cost.
- Normative gap indices for 29 municipalities, revealing cross-municipal asymmetries in regulatory attention—including the absence of social housing provisions in five municipalities—that reflect the socioeconomic stratification of the Gran Santiago metropolitan area.
- An empirical baseline for small VLM classification on specialized legal text, documenting Moondream's strengths (visual document understanding, map interpretation) and limitations (text classification without fine-tuning, high attrition on complex articles) with direct implications for method selection in urban regulatory research.

The study contributes at three levels. Methodologically, it provides a fully open, reproducible pipeline for PRC analysis that any research group can deploy without cloud API access. Empirically, it extends the normative gap analysis of the Gran Santiago PRC corpus with a methodology that is transparent, auditable, and replicable. Critically, it positions the computational accessibility of GeoAI tools as a political question—one whose answers have consequences for the geographic distribution of urban knowledge production.

Implications for planning practice. Beyond academic research, the pipeline described here has direct operational applications for municipal and metropolitan planning offices working under fiscal and digital-infrastructure constraints. A planning department in a typical Gran Santiago municipality—which often operates with limited GIS staff, no in-house AI capacity, and modest computing budgets—can deploy the same Moondream and keyword pipeline on a single mid-range laptop to: (i) generate consistent thematic indices of its own PRC and benchmark them against neighboring municipalities; (ii) automatically extract regulatory metadata from incoming or legacy PDFs for archival and public consultation purposes; (iii) screen draft PRC modifications for thematic coverage gaps prior to public participation processes; and (iv) populate municipal open-data portals with structured indicators derived from previously unstructured regulatory text. Because the pipeline runs entirely on local hardware, none of these uses requires transmitting poten-

tially sensitive municipal data to external servers, nor does it incur recurrent subscription costs. This combination—auditable methods, zero marginal cost, full data sovereignty—can serve as the operational substrate on which more ambitious digital-planning workflows are incrementally built, particularly in Latin American jurisdictions where the gap between regulatory ambition and computational capacity is widest.

Future extensions include: (a) fine-tuning Moondream or a similarly-sized model on labeled Chilean PRC data to improve text classification performance; (b) evaluating additional small VLMs (LLaVA, SmolVLM, Phi-3-Vision) to establish broader baselines for the model class; (c) incorporating human-annotated gold standard validation to complement the keyword silver standard; (d) extending the pipeline to the national Chilean PRC corpus (346 municipalities); and (e) developing a standardized API for PRC knowledge extraction that could serve municipal and regional planning agencies directly.

The cities of the Global South are not passive objects of data extraction—they are active sites of regulatory production, urban knowledge, and territorial imagination. Making the tools that read these cities accessible to the researchers who live in them is a condition for a more geographically equitable urban science.

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Data Availability Statement: The PRC corpus is available from Chilean municipal and regional planning archives under public access provisions of Chilean Law 20.285 (Ley de Acceso a la Informacion Publica).

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Conflicts of Interest: The author declares no conflicts of interest.

Appendix A. Moondream Prompts (In Spanish)

Classification prompt:
Clasifica este texto de una ordenanza urbana chilena.
Ejes posibles: zoning, land_use, mobility, participation, sustainability, social_housing.
Texto: {text}
Responde solo con lista JSON, ejemplo: [“zoning”]. Si ninguno aplica: []
Map interpretation prompt:
Eres un experto en cartografia urbana chilena.
Describe brevemente el mapa o plano urbano que aparece en esta imagen:

- Tipo de mapa (uso de suelo, zonificacion, vialidad, etc.)
- Municipio o area representada si es identificable
- Leyenda principal o categorias visibles
- Responde en espanol, maximo 3 oraciones.

Classification prompt in English:
Classify this text from a Chilean urban ordinance.
Possible axes: zoning, land use, mobility, participation, sustainability, social housing.
Text: {text}
Respond only with a JSON list, for example: [“zoning”]. If none apply: []
Map interpretation prompt:
You are an expert in Chilean urban cartography.
Briefly describe the urban map or plan shown in this image:

- Type of map (land use, zoning, road network, etc.)
- Municipality or area represented if identifiable
- Main legend or visible categories
- Respond in English, maximum 3 sentences.

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